

CS 419

Evaluation of IR Systems

Why System Evaluation?

- There are many retrieval models/ algorithms/ systems, which one is the best?
- What is the best component for:
 - Ranking function (dot-product, cosine, ...)
 - Term selection (stopword removal, stemming...)
 - Term weighting (TF, TF-IDF,...)
- How far down the ranked list will a user need to look to find some/all relevant documents?

Difficulties in Evaluating IR Systems

- Effectiveness is related to the **relevancy** of retrieved items.
- Relevancy is not typically binary but continuous.
- Even if relevancy is binary, it can be a difficult judgment to make.

Difficulties in Evaluating IR Systems

- Relevancy, from a human standpoint, is:
 - **Subjective**: depends upon a specific user's judgment.
 - **Situational**: relates to user's current needs.
 - **Cognitive**: depends on human perception and behavior.
 - **Dynamic**: changes over time.

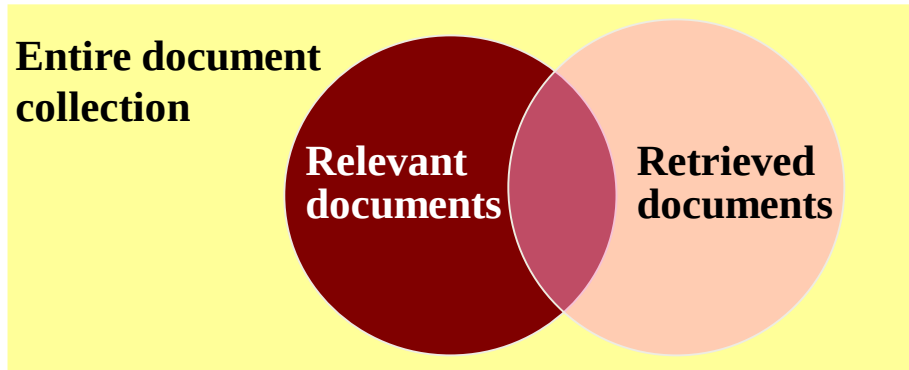
Human Labeled Corpora (Gold Standard)

- Start with a corpus of documents.
- Collect a set of queries for this corpus.
- Have one or more human experts exhaustively label the relevant documents for each query.
- Typically assumes binary relevance judgments.
- Requires considerable human effort for large document/query corpora.

Note

- The **information need** is translated into a **query**
- Relevance is assessed relative to the **information need**, *not* the **query**
 - E.g., Information need: *I'm looking for information on whether drinking red wine is more effective at reducing heart attack risks than white wine.*
 - Query: ***wine red white heart attack effective***
- Evaluate whether the doc addresses the information need, not whether it has these words

Precision and Recall



irrelevant	retrieved & irrelevant	Not retrieved & irrelevant
	retrieved & relevant	not retrieved but relevant
relevant	retrieved	not retrieved

$$\text{recall} = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of relevant documents}}$$

$$\text{precision} = \frac{\text{Number of relevant documents retrieved}}{\text{Total number of documents retrieved}}$$

Precision and Recall

- Precision
 - The ability to retrieve top-ranked documents that are mostly relevant.
- Recall
 - The ability of the search to find ***all*** of the relevant items in the corpus.

Determining Recall is Difficult

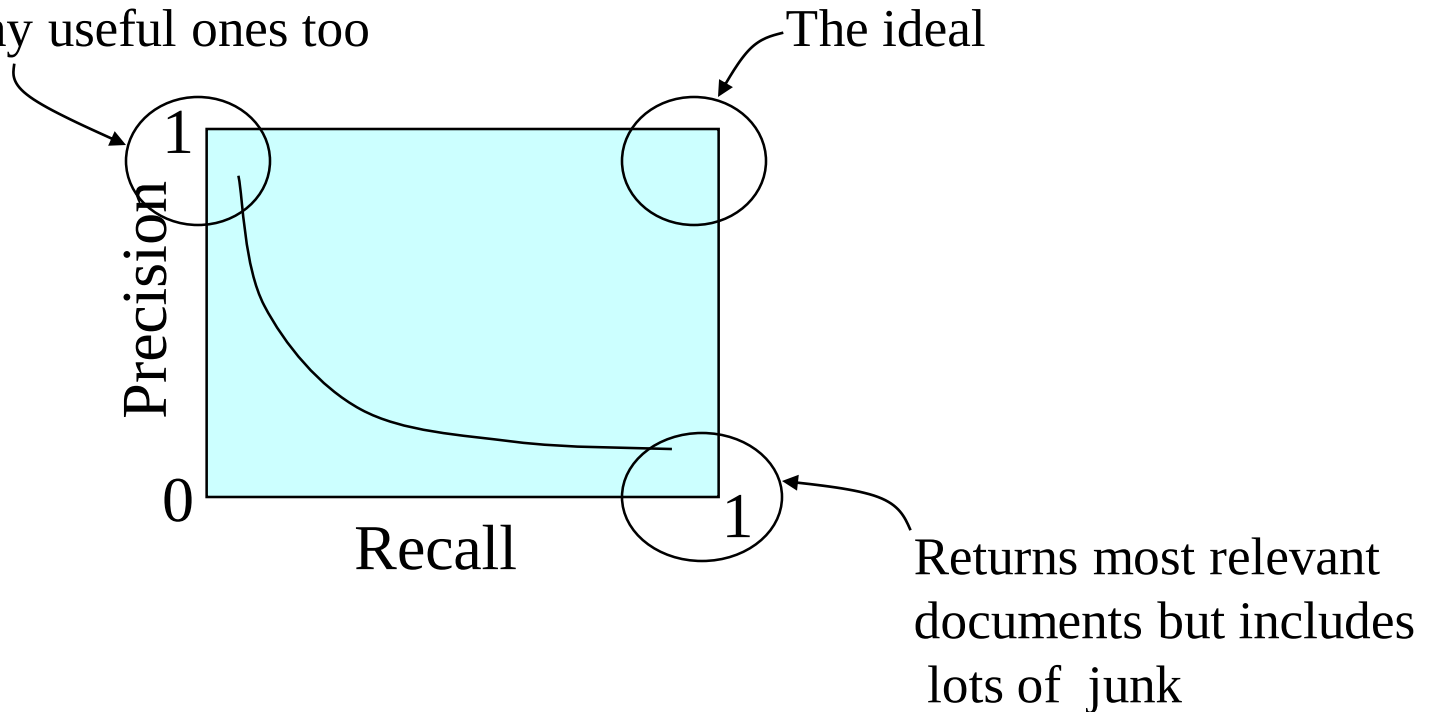
- Total number of relevant items is sometimes not available:
 - Sample across the database and perform relevance judgment on these items.
 - Apply different retrieval algorithms to the same database for the same query. The aggregate of relevant items is taken as the total relevant set.

Precision/Recall

- You can get high recall (but low precision) by retrieving all docs for all queries!
- Recall is a non-decreasing function of the number of docs retrieved
- In a good system, precision decreases as either the number of docs retrieved or recall increases
 - This is not a theorem, but a result with strong empirical confirmation

Trade-off between Recall and Precision

Returns relevant documents but misses many useful ones too



Computing Recall/Precision Points

- For a given query, produce the ranked list of retrievals.
- Adjusting a threshold on this ranked list produces different sets of retrieved documents, and therefore different recall/precision measures.
- Mark each document in the ranked list that is relevant according to the gold standard.
- Compute a recall/precision pair for each position in the ranked list that contains a relevant document.

Computing Recall/Precision Points:

Example 1

n	doc #	relevant
1	588	x
2	589	x
3	576	
4	590	x
5	986	
6	592	x
7	984	
8	988	
9	578	
10	985	
11	103	
12	591	
13	772	x
14	990	

Let total # of relevant docs = 6
Check each new recall point:

$R=1/6=0.167$; $P=1/1=1$

$R=2/6=0.333$; $P=2/2=1$

$R=3/6=0.5$; $P=3/4=0.75$

$R=4/6=0.667$; $P=4/6=0.667$

$R=5/6=0.833$; $p=5/13=0.38$

Missing one
relevant document.
Never reach
100% recall

Computing Recall/Precision Points:

Example 2

n	doc #	relevant
1	588	x
2	576	
3	589	x
4	342	
5	590	x
6	717	
7	984	
8	772	x
9	321	x
10	498	
11	113	
12	628	
13	772	
14	592	x

Let total # of relevant docs = 6
Check each new recall point:

$R=1/6=0.167$; $P=1/1=1$

$R=2/6=0.333$; $P=2/3=0.667$

$R=3/6=0.5$; $P=3/5=0.6$

$R=4/6=0.667$; $P=4/8=0.5$

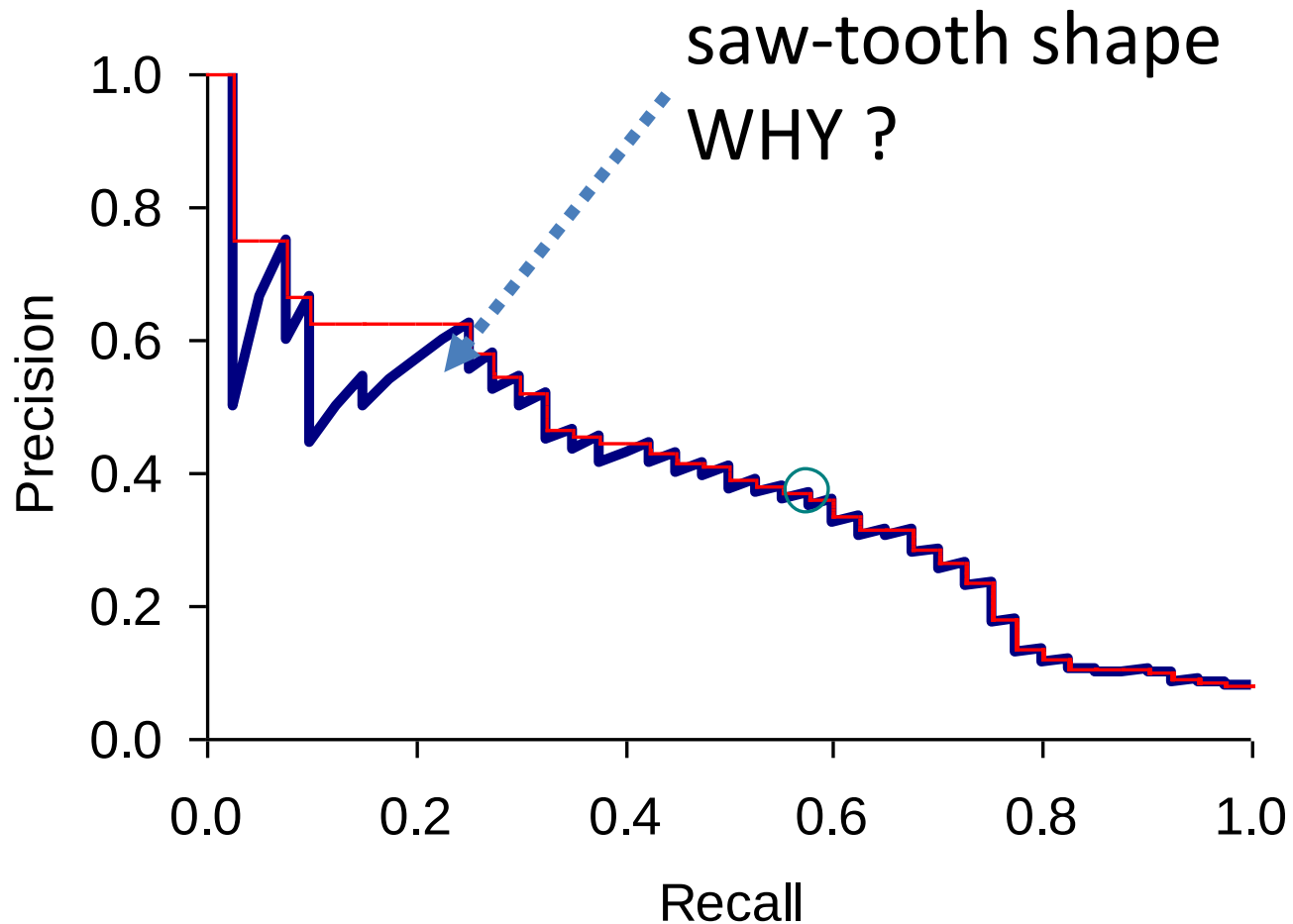
$R=5/6=0.833$; $P=5/9=0.556$

$R=6/6=1.0$; $p=6/14=0.429$

Evaluating ranked results

- Evaluation of ranked results:
 - The system can return any number of results
 - By taking various numbers of the top returned documents (levels of recall), the evaluator can produce a *precision-recall curve*

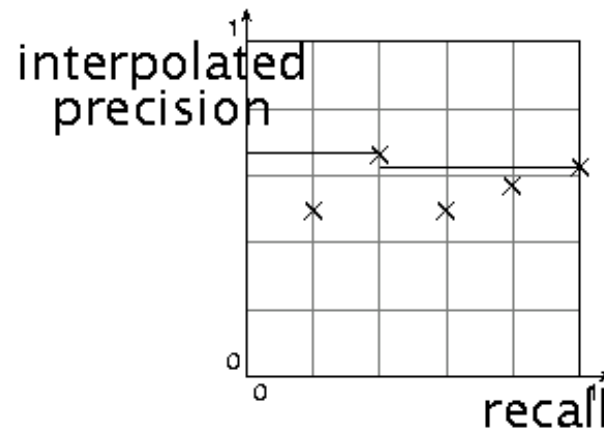
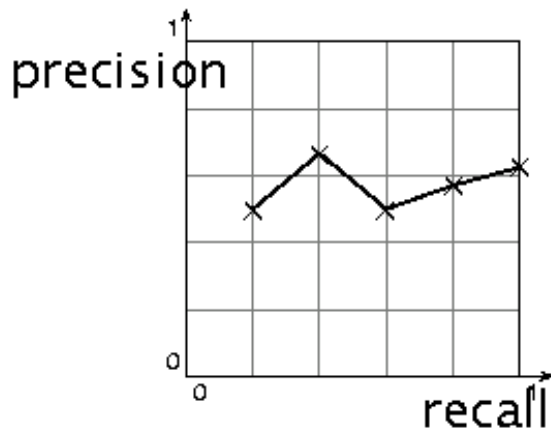
A precision-recall curve



Interpolated precision

- TO: remove the jiggles (i.e. “tooth”)
- The interpolated precision at a certain recall level r is defined as the highest precision found for any recall level $r' \geq r$:

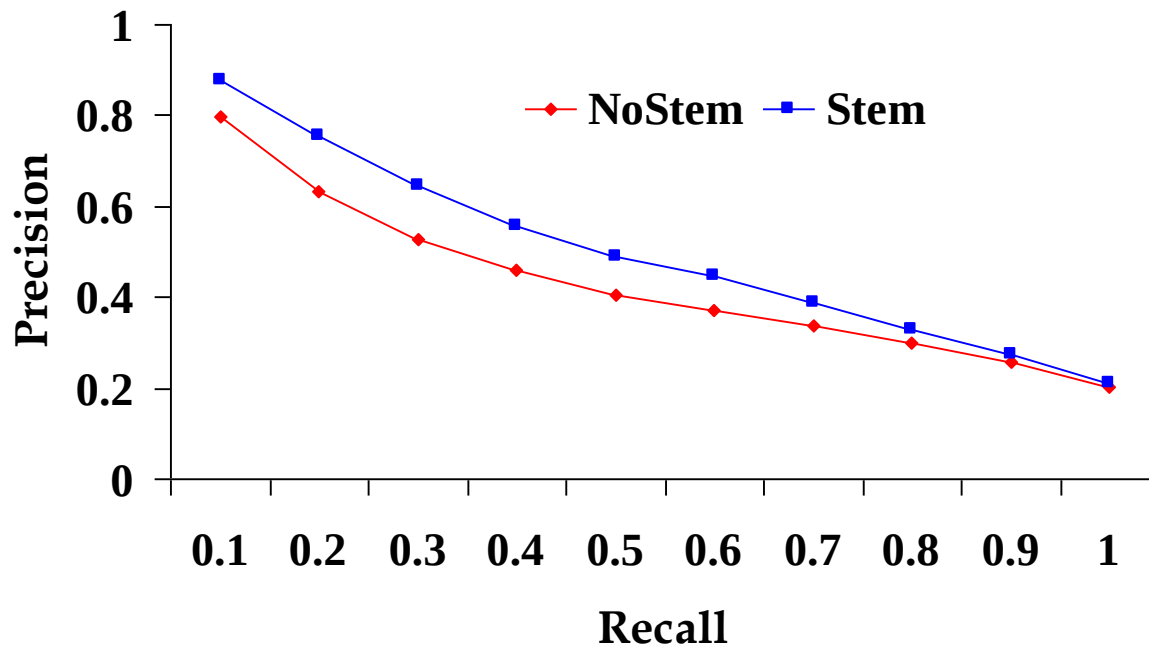
$$P(r) = \max_{r' \geq r} P(r')$$



- Graphs are good, but people want summary measures!
 - Precision at fixed retrieval level
 - Precision-at- k : Precision of top k results
 - Perhaps appropriate for most of web search: all people want are good matches on the first one or two results pages
 - But: averages badly and has an arbitrary parameter of k
 - 11-point interpolated average precision
 - The standard measure in the early TREC competitions: you take the precision at 11 levels of recall varying from 0 to 1 by tenths of the documents, using interpolation (the value for 0 is always interpolated!), and average them
 - Evaluates performance at all recall levels

Compare Two or More Systems

- The curve closest to the upper right-hand corner of the graph indicates the best performance



Difficulties in using precision/recall

- Need human relevance assessments
 - People aren't reliable assessors
- Assessments have to be binary
 - Nuanced assessments?
- Heavily skewed by collection/authorship
 - Results may not translate from one domain to another

R- Precision

- Precision at the R-th position in the ranking of results for a query that has R relevant documents.

n	doc #	relevant
1	588	x
2	589	x
3	576	
4	590	x
5	986	
6	592	x
7	984	
8	988	
9	578	
10	985	
11	103	
12	591	
13	772	x
14	990	

R = # of relevant docs = 6

R-Precision = $4/6 = 0.67$

F-Measure

- One measure of performance that takes into account both recall and precision.
- Harmonic mean of recall and precision:

$$F = \frac{2PR}{P + R} = \frac{2}{\frac{1}{R} + \frac{1}{P}}$$

- Compared to arithmetic mean, both need to be high for harmonic mean to be high.

Parameterized F Measure

- A variant of F measure that allows weighting emphasis on precision over recall:

$$E = \frac{(1 + \beta^2)PR}{\beta^2 P + R} = \frac{(1 + \beta^2)}{\frac{\beta^2}{R} + \frac{1}{P}}$$

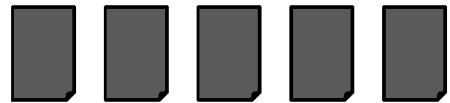
- Value of β controls trade-off:
 - $\beta = 1$: Equally weight precision and recall ($E=F$).
 - $\beta > 1$: Weight recall more.
 - $\beta < 1$: Weight precision more.

Mean Average Precision (MAP)

- **Average Precision:** Average of the precision values at the points at which each relevant document is retrieved.
- **Mean Average Precision:** Average of the average precision value for a set of queries.

Mean Average Precision (MAP)

- Given that n docs are retrieved
 - Compute the precision (at rank) where each (new) relevant document is retrieved $\Rightarrow p(1), \dots, p(k)$, if we have k rel. docs
 - E.g., if the first rel. doc is at the 2nd rank, then $p(1)=1/2$.
 - If a relevant document never gets retrieved, we assume the precision corresponding to that rel. doc to be zero
- Compute the average over all the relevant documents
 - Average precision = $(p(1)+\dots p(k))/k$

 = relevant documents for query 1

Ranking #1 

Recall
Precision

 = relevant documents for query 2

Ranking #2 

Recall
Precision

average precision query 1

average precision query 2

mean average precision =

Mean Average Precision (MAP)

- Summarize rankings from multiple queries
- Gives us (non-interpolated) average precision, which captures both precision and recall and is sensitive to the rank of each relevant document
- most commonly used measure in research papers
- assumes user is interested in finding many relevant documents for each query
- requires many relevance judgments in text collection

Non-Binary Relevance

- Documents are rarely entirely relevant or non-relevant to a query
- Many sources of *graded relevance judgments*
 - E.g. Relevance judgments on a 5-point scale

Cumulative Gain

- With graded relevance judgments, we can compute the *gain* at each rank.
- **Cumulative Gain** at rank n :

$$CG_n = \sum_{i=1}^n rel_i$$

(Where rel_i is the graded relevance of the document at position i)

n	doc #	relevance	CG _n
		(gain)	
1	588	1.0	1.0
2	589	0.6	1.6
3	576	0.0	1.6
4	590	0.8	2.4
5	986	0.0	2.4
6	592	1.0	3.4
7	984	0.0	3.4
8	988	0.0	3.4
9	578	0.0	3.4
10	985	0.0	3.4
11	103	0.0	3.4
12	591	0.0	3.4
13	772	0.2	3.6
14	990	0.0	3.6

Discounting Based on Position

- Users care more about high-ranked documents, so we **discount** results by $1/\log_2(rank)$

- Discounted Cumulative Gain:**

$$DCG_n = rel_1 + \sum_{i=2}^n \frac{rel_i}{\log_2 i}$$

n	doc #	rel (gain)	CG _n	log _n	DCG _n
1	588	1.0	1.0	-	1.00
2	589	0.6	1.6	1.00	1.60
3	576	0.0	1.6	1.58	1.60
4	590	0.8	2.4	2.00	2.00
5	986	0.0	2.4	2.32	2.00
6	592	1.0	3.4	2.58	2.39
7	984	0.0	3.4	2.81	2.39
8	988	0.0	3.4	3.00	2.39
9	578	0.0	3.4	3.17	2.39
10	985	0.0	3.4	3.32	2.39
11	103	0.0	3.4	3.46	2.39
12	591	0.0	3.4	3.58	2.39
13	772	0.2	3.6	3.70	2.44
14	990	0.0	3.6	3.81	2.44

Normalized Discounted Cumulative Gain (NDCG)

- To compare DCGs, normalize values so that a *ideal ranking* would have a **Normalized DCG** of 1.0
- Normalized Cumulative Gain (NDCG) at rank n
 - Normalize DCG at rank n by the DCG value at rank n of the ideal ranking
 - The ideal ranking would first return the documents with the highest relevance level, then the next highest relevance level, etc
 - Compute the precision (at rank) where each (new) relevant document is retrieved $\Rightarrow p(1), \dots, p(k)$, if we have k rel. docs

Normalized Discounted Cumulative Gain (NDCG)

- To compare DCGs, normalize values so that a *ideal ranking* would have a **Normalized DCG** of 1.0
- Ideal ranking:

n	doc #	rel (gain)	CG _n	log _n	DCG _n
1	588	1.0	1.0	0.00	1.00
2	589	0.6	1.6	1.00	1.60
3	576	0.0	1.6	1.58	1.60
4	590	0.8	2.4	2.00	2.00
5	986	0.0	2.4	2.32	2.00
6	592	1.0	3.4	2.58	2.39
7	984	0.0	3.4	2.81	2.39
8	988	0.0	3.4	3.00	2.39
9	578	0.0	3.4	3.17	2.39
10	985	0.0	3.4	3.32	2.39
11	103	0.0	3.4	3.46	2.39
12	591	0.0	3.4	3.58	2.39
13	772	0.2	3.6	3.70	2.44
14	990	0.0	3.6	3.81	2.44



n	doc #	rel (gain)	CG _n	log _n	IDCG _n
1	588	1.0	1.0	0.00	1.00
2	592	1.0	2.0	1.00	2.00
3	590	0.8	2.8	1.58	2.50
4	589	0.6	3.4	2.00	2.80
5	772	0.2	3.6	2.32	2.89
6	576	0.0	3.6	2.58	2.89
7	986	0.0	3.6	2.81	2.89
8	984	0.0	3.6	3.00	2.89
9	988	0.0	3.6	3.17	2.89
10	578	0.0	3.6	3.32	2.89
11	985	0.0	3.6	3.46	2.89
12	103	0.0	3.6	3.58	2.89
13	591	0.0	3.6	3.70	2.89
14	990	0.0	3.6	3.81	2.89

Normalized Discounted Cumulative Gain (NDCG)

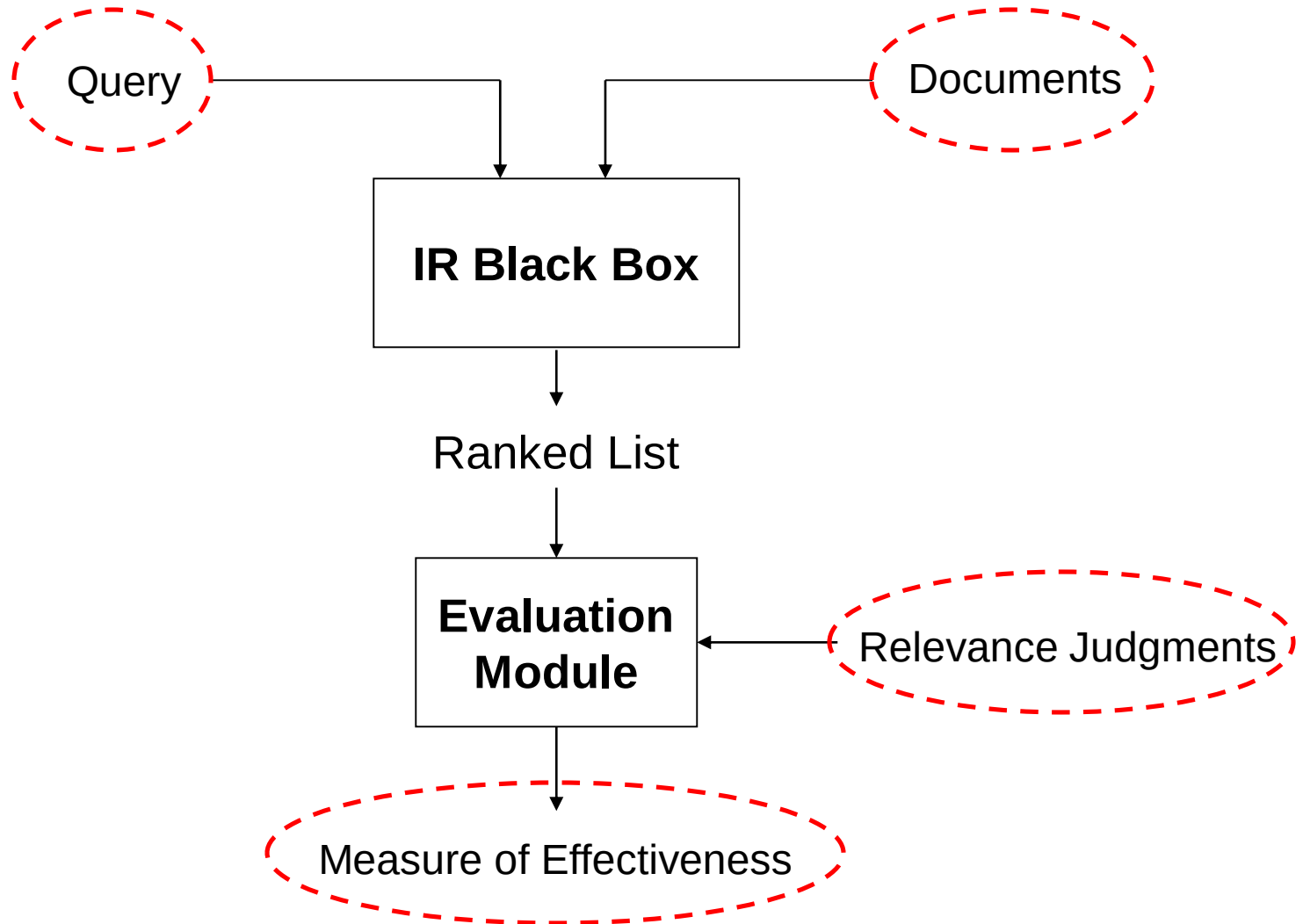
- Normalize by DCG of the ideal ranking:

$$NDCG_n = \frac{DCG_n}{IDCG_n}$$

- $NDCG \leq 1$ at all ranks
- NDCG is comparable across different queries
- NDCG is now quite popular in evaluating Web search

n	doc #	rel	DCG_n	$IDCG_n$	$NDCG_n$
		(gain)			
1	588	1.0	1.00	1.00	1.00
2	589	0.6	1.60	2.00	0.80
3	576	0.0	1.60	2.50	0.64
4	590	0.8	2.00	2.80	0.71
5	986	0.0	2.00	2.89	0.69
6	592	1.0	2.39	2.89	0.83
7	984	0.0	2.39	2.89	0.83
8	988	0.0	2.39	2.89	0.83
9	578	0.0	2.39	2.89	0.83
10	985	0.0	2.39	2.89	0.83
11	103	0.0	2.39	2.89	0.83
12	591	0.0	2.39	2.89	0.83
13	772	0.2	2.44	2.89	0.84
14	990	0.0	2.44	2.89	0.84

Automatic Evaluation



These are the four things we need!

Evaluation at large search engines

- Recall is difficult to measure on the web
- Search engines often use precision at top k , e.g., $k = 10$
- ... or measures that reward you more for getting rank 1 right than for getting rank 10 right.
 - NDCG (Normalized Cumulative Discounted Gain)
- Search engines also use non-relevance-based measures.
 - Clickthrough on first result
 - Not very reliable if you look at a single clickthrough ... but pretty reliable in the aggregate.

Discussion

- Relevance vs Marginal Relevance
 - A document can be redundant even if it is highly relevant
 - Duplicates
 - The same information from different sources
 - Marginal relevance is a better measure of utility for the user ?
- Do later documents in the ranking add new information beyond what is already given in higher documents ?
 - Choice of retrieved set should encourage **diversity** and **novelty**.

What you should know

- Measures:
 - Precision/Accuracy, Recall
 - Precision and Recall curve
 - R-precision
 - F-measure, MAP
 - NCDG, non-binary relevance
- Pros and Cons

Refs/Slide Credit

- CS276 – Stanford University